

Netrokona University
Department of Computer Science and Engineering
Midterm, 3rd Year 1st Semester-2024
Course Code: CSE 3104 **Course Title: Theory of Computation Automata**
Full Marks: 20 **Time: 1 Hour**

[The values on the right-hand side indicate marks allocated for that question only.]

1. ~~(a)~~ Define Alphabet, String and Language with examples. 3

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Solution:

In theoretical computer science, these terms define how computational models process information:

- **Alphabet (Σ):** A finite, non-empty set of symbols.
- **Example:** The binary alphabet ($\Sigma = \{0, 1\}$). 1
- **String:** A finite sequence of symbols taken from a specific alphabet.
- **Example:** Over the binary alphabet, "0110" is a valid string. 1
- **Language:** A set of strings constructed from a given alphabet.
- **Example:** The set of all strings that start with 'a' over the alphabet ($\Sigma = \{a, b\}$). 1, 2

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- (b) Create DFA accepting for the following languages over the alphabet $\{0, 1\}$
- i. Construct a DFA over $\{0, 1\}$ that accepts all strings where the number of 0's is divisible by 3. 2
 - ii. The set of strings that either begin or end (or both) with 01. .
- ~~(c)~~ The set of strings beginning with a 1 that, when interpreted as a binary integer, is a multiple of 5. For example, string 101, 1010, are in the language and 1111, 0, 100, 111 are not. 4
- ~~(d)~~ Explain why NFA is not more powerful than DFA, although it looks more general. 1
2. ~~(a)~~ Briefly explain Non-deterministic Finite Automata. 2
- ~~(b)~~
- i. Construct an NFA for the language of strings over $\{a, b\}$ where the string starts and ends with the same symbol. 3
 - ii. Build an NFA for strings over $\{0, 1\}$ that end with either 01 or 10.
- ~~(c)~~
- i. Write a Regular Expression for the set of all strings over $\{0, 1\}$ that contain exactly one occurrence of 00. 2
 - ii. Give an RE over $\{0, 1\}$ for all strings where the number of 1's is odd.

(d) Convert the following DFA into Regular expression.

3

